IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 9 June 2025, accepted 11 July 2025, date of publication 14 July 2025, date of current version 21 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3589010

RESEARCH ARTICLE

YOLOv11-ND: A Method for Identifying Traffic Targets in Nighttime Urban Environments

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ABSTRACT Traffic target recognition is a crucial technology that has drawn a lot of interest due to the quick development of unmanned and assisted driving systems. However, the precision and performance of target recognition for the more complicated nighttime environment are lower, and the majority of the present research on traffic target recognition concentrates on the daytime. By using nighttime traffic targets as the research object, this paper suggests YOLOv11-ND, an enhanced target recognition method, to address the aforementioned issues. First, based on WTConv, the WTC3k2 module is intended to take the position of C3k2 in the backbone part. This reduces the amount of model parameters without sacrificing precision. Then, to boost the fusion ability of multi-scale features, the HS-FPN network structure is adopted in the neck section. This improves the detection performance. Lastly, the model is optimized using the Focaler-GIoU loss function to further enhance the detection performance. In comparison to the baseline model YOLOv11, the enhanced model YOLOv11-ND improved the P, R, mAP50, and mAP50-95 measures by 3.1%, 3.4%, 4%, and 3.6%, respectively, according to experimental validation using the FLIR dataset.The method can successfully increase the precision of traffic target detection in urban settings at night, according to the testing results.

INDEX TERMS FLIR, Focaler-GIoU, HS-FPN, night target detection, WTC3k2, YOLOv11.

I. INTRODUCTION

The number of national automobiles has been rising in tandem with the acceleration of urbanization and the ongoing improvement of the social and economic level. The constant increase in traffic volume raises the requirements for both traffic management and safety. Traditional visual recognition methods have high recognition precision in well-lit conditions but have difficulty accurately recognizing and tracking traffic targets at night. Statistics show that a significant percentage of all traffic accidents in recent years have occurred at night. In order to ensure traffic safety, research has thus turned to traffic target recognition techniques for metropolitan nighttime situations [1].

Traditional infrared object identification methods [2], [3], [4], [5] typically have drawbacks, including low recognition precision and processing complexity, which make it difficult to achieve popularity. Two-stage recognition algorithms, exemplified by R-CNN [6], Fast R-CNN [7], and Faster R-CNN [8], have been progressively used in the field of traffic target recognition in recent years due to the rapid advancement of deep learning. However, the two-stage recognition algorithm is not suitable for real-time applications due to its limitations in detection speed. Until the emergence of the single-stage recognition algorithm YOLO [9], [10], [11], which significantly improved the detection speed of the model, deep learning has been more widely used in the field of traffic target detection. Nevertheless, current detection algorithms continue to struggle to achieve precise detection because of the poor nighttime visibility of metropolitan locations.

In this paper, an enhanced YOLOv11 algorithm (i.e., YOLOv11-ND) is proposed to address the shortcomings of YOLOv11 in recognizing traffic targets in nighttime urban

The associate editor coordinating the review of this manuscript and approving it for publication was Antonio J. R. Neves.

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environments. The main contributions of the algorithm are as follows:

(1) In the backbone section, the WTC3k2 module based on WTConv replaces the original C3k2 module. This results in model lightweighting and enhanced detection performance by reducing the number of parameters without sacrificing feature information;

(2) The original neck part is replaced by the HS-FPN(High-level Screening-feature Fusion Pyramid Network) network structure, which effectively combines low-scale feature detail information with high-level feature semantic information. This improves the model's capacity to handle the fusion and extraction of multi-scale features, which in turn increases the detection model's precision and resilience;

(3) The Focaler-GIoU loss function is introduced to improve the detection performance in different detection tasks by focusing on different regression samples. The effect of the distribution of simple and difficult samples in the bounding box regression on the regression is also considered, and the geometric relationship between the bounding boxes is utilized to further improve the performance of the model.

II. RELATED WORK

With the acceleration of urbanization, the number of people traveling at night is also increasing [12]. However, insufficient light and low road visibility at night lead to frequent traffic accidents. Therefore, accurately identifying various types of traffic targets in nighttime urban environments is of great significance in reducing traffic accidents and improving traffic safety.

Traditional infrared object detection obtains target information from preprocessed images and then realizes target detection and tracking. The three primary components of image preprocessing are non-uniformity correction, denoising, and image enhancement. Tao et al. [13] proposed an adaptive algorithm for low-illumination images based on the traditional Retinex theory and used a bilateral filter to recover the detail information of the picture, thus enhancing the acquisition of effective information. Xu et al. [14] used a median filter for initial noise reduction of the image and then used a compression sensor to achieve the denoising effect so that the signal-to-noise ratio of the image is significantly improved. Huang et al. [15] proposed an adaptive multi-point calibration correction algorithm, which utilizes the residual difference as a judgment condition when selecting the calibration point to realize adaptive determination of the calibration point and then improve the calibration effect of the infrared image.

Scale Invariant Feature Transform (SIFT) [16], Aggregated Channel Feature (ACF) [17], Histogram of Oriented Gradients (HOG) [18], and other manually created feature extractors are typically used in traditional infrared detection techniques to extract the target's features. Classifiers like Adaboost [19] and Support Vector Machine (SVM) [20] are used to categorize the targets. However, it is challenging to gain traction due to the high processing costs and poor generalization capabilities of conventional infrared object detection methods.

Until the emergence of deep learning methods, VIT [21], [22], [23] and YOLO [24], [25], [26] were widely used for nighttime traffic target recognition due to their advantages of high detection precision and fast efficiency. In order to lighten the night detection model, Huang et al. [27] proposed a night detection method, M-YOLO, combining MobileNet v2 and YOLO v3 to lighten the night detection model. Li et al. [28] added ResNet and DCGAN networks to YOLOv5 to improve target recognition at night. Li et al. [29] suggested the YOLO-FIR infrared image detection network based on YOLOV5 to address the issues of low recognition precision and high false detection rate of small targets at night. This network successfully resolves the aforementioned issues. Bowen et al. [30] presented a YOLOv7-based night vision target detection algorithm that decreases the number of parameters and enhances the model's resilience and generalization capabilities by first preprocessing the image using gamma transform and variance equalization and then optimizing YOLOv7 using the GSConv, ShuffleNetv2, and SIoU modules. Ren [31] considered the effect of low illumination, smoke, fog, and other factors on the recognition precision and integrated a new module, SPD, in YOLOv8 to realize the improvement of low-resolution image recognition precision.

Computational efficiency in industrial deployments remains a primary limitation of deep learning-based detection models. To address this challenge, lightweight feature extraction networks such as EfficientDet [32], MobileNet [33], and NanoDet [34] have been adopted as backbones to optimize detectors like YOLO and DETR. However, this approach to improvement typically comes at the expense of precision degradation.

To enhance detection precision for nighttime targets, numerous researchers employ image preprocessing techniques. These include amplifying the visual characteristics of objects, performing illumination estimation, and applying image enhancement alongside dehazing. In order to adapt to object detection under different low-light conditions. Yin et al. [35] proposed a detail processing module (DPM) to enhance the detail of images, which consists of context branch and edge branch. Cai et al. [36] formulated a simple yet principled One-stage Retinex-based Framework (ORF). ORF first estimates the illumination information to light up the low-light image and then restores the corruption to produce the enhanced image. And designed an Illumination-Guided Transformer (IGT) that utilizes illumination representations to direct the modeling of non-local interactions of regions with different lighting conditions to enhance the performance of the low-light object detection.

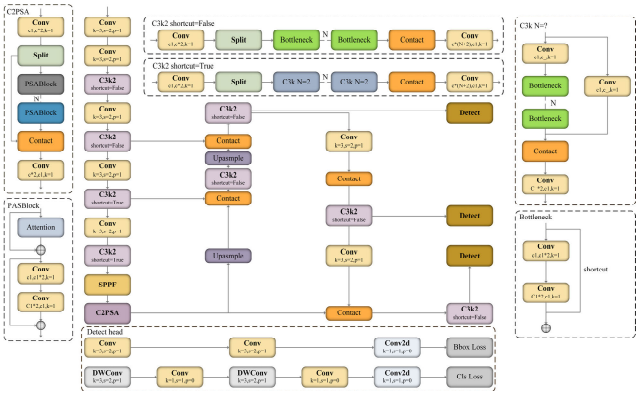


FIGURE 1. Network structure of YOLOV11.

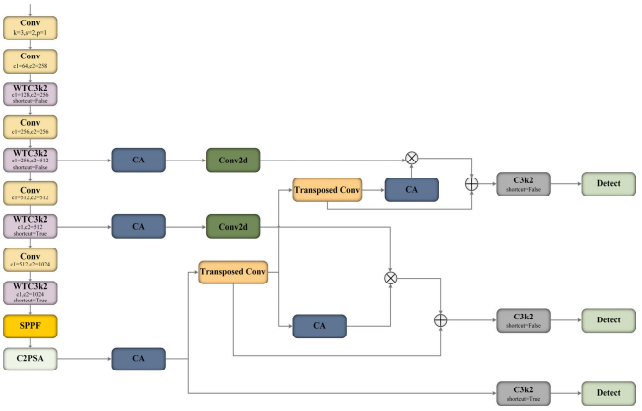


FIGURE 2. Structure of YOLOV11-ND.

III. METHOD

A. YOLOV11 NETWORK STRUCTURE

Fig. 1 depicts the network structure of YOLOv11, the most recent model in the YOLO series, which consists of three primary parts: the backbone, neck, and head. In order to increase computational efficiency and speed up feature extraction in image processing, the YOLOv11 backbone replaces the original C2f module with the C3k2 module. The C2PSA module is added to the feature enhancement network section to improve the efficiency of extracting important features from pictures. In addition, YOLOv11 utilizes a more efficient network architecture and training process. The depth-separable method used by the detection head portion minimizes unnecessary calculations and streamlines the computing process, significantly lowering the model's computation and parameter count. YOLOv11 improves the sensitivity and precision of model detection by refining the architecture mentioned above.

B. IMPROVED YOLOV11 MODEL FOR NIGHT TRAFFIC TARGET DETECTION

In order to address the issue of low visibility and recognition challenges in urban environments at night, this paper suggests a night detection model based on Yolov11—YOLOV11-ND. Fig. 2 depicts the model network structure. First, the original C3k2 plate in the model's backbone is replaced with the WTC3k2, which lowers the number of parameters while maintaining precision and achieving the model's lightweight. The neck section then employs the HS-FPN network structure to efficiently integrate the low-scale feature detail information with the high-level feature semantic information, enhancing the model's feature extraction and detection precision. Finally, the Focaler-GIoU loss function is introduced to improve the detection performance in different detection tasks by focusing on different regression samples. And the geometric relationship between the bounding boxes is utilized to further improve the detection performance of the model.

C. WTC3K2

This paper introduces the WTC3k2 module based on the architecture of WTConv [37] in the backbone part, which reduces the number of parameters while maintaining the precision of model detection. Fig. 3 depicts the construction of the WTC3k2 network.

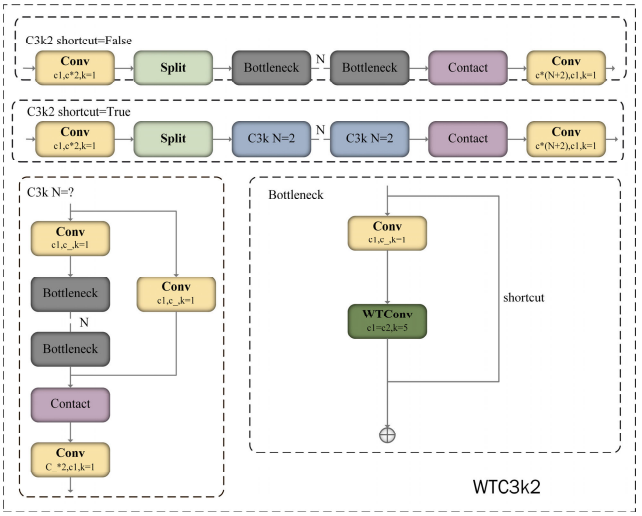


FIGURE 3. Structure of WTC3k2.

Where Fig. 5 depicts the WTConv network structure. Firstly, the input image X is subjected to a wavelet downsampling operation for image feature extraction, which generates a low-frequency map X_{LL} as well as three high-frequency maps X_{LH}, X_{HL}, and X_{HH}, whose computational process is shown in equations (1) and (2).

$$\begin{cases} f_{LL} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\ f_{LH} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \\ f_{HL} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} \\ f_{HH} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \end{cases} \tag{1}$$

where f_{LL} is a low-pass filter and f_{LH}, f_{HL}, and f_{HH} are high-pass filters.

A deep convolution with a step size of 2 is performed through the above four filters. The output of the convolution is shown in equation (2).

$$[X_{LL}, X_{LH}, X_{HL}, X_{HH}] = Conv\left([f_{LL}, f_{LH}, f_{HL}, f_{HH}], X\right) \tag{2}$$

Then the low-frequency map is continuously decomposed to achieve the effect of improving the corresponding response of the convolutional layer, and the decomposition process is shown in equation (3):

$$X_{LL}^{(i)}, X_H^{(i)} = WT\left(X_{LL}^{(i-1)}\right) \tag{3}$$

where i represents the layer level, and when i=1, there is $X_{LL}^{(i-1)} = X$. $\boldsymbol{X_{LL}^{(i)}}$ and $X_H^{(i)}$ are the low-frequency maps and all the high-frequency maps in layer i, respectively, and $X_H^{(i)}$ is composed of $X_{LH}^{(i)}$, $X_{HL}^{(i)}$ *and* $X_{HH}^{(i)}$.

Next, small kernel depth convolution is performed on different frequency maps as shown in equation (4):

$$Y_{LL}^{(i)}, Y_H^{(i)} = Conv\left(W^{(i)}, (X_{LL}^{(i)}, X_H^{(i)})\right) \tag{4}$$

where $W^{(i)}$ is the weight tensor of the $K \times K$ depth kernel with four times as many input channels as the input tensor. This separates the convolutions between the different frequency components and allows the smaller convolution kernels to operate in a larger region of the original input, thus increasing the receptive field of the input. Specific instructions are shown in Fig. 4. In this example, a 3 × 3 convolution of the lower frequency bands of the secondary wavelet domain $X_{LL}^{(2)}$ is performed, resulting in a 9-parameter convolution in response to the lower frequencies of the 12 × 12 receptive fields in the input X.

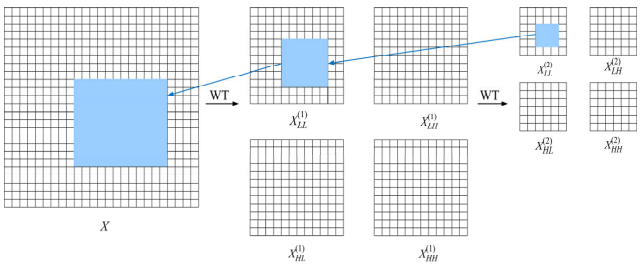


FIGURE 4. Convolution in the wavelet domain produces a larger sensory field.

Finally, the outputs of different frequencies are combined using IWT (Inverse Wavelet Transform) to obtain the sum $Z^{(i)}$ of the output of the ith layer. The calculation procedure is shown in equation (5):

$$Z^{(i)} = IWT\left(Y_{LL}^{(i)} + Z^{(i+1)}, Y_H^{(i)}\right) \tag{5}$$

WTC3k2 utilizes the wavelet transform to achieve an increase in the receptive field through multi-frequency response, enabling the model to capture low-frequency information over a wider range and enhancing the CNN's ability to respond to shapes rather than textures while avoiding over-parameterization.

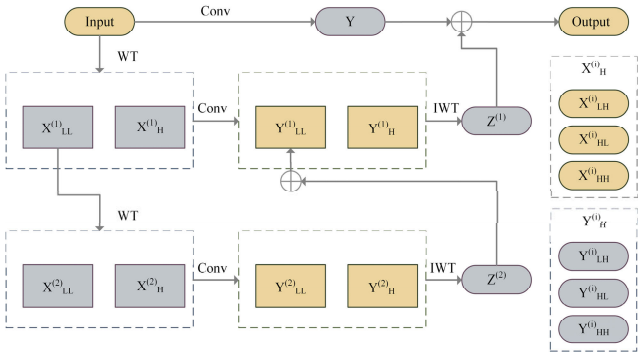


FIGURE 5. Structure of WTConv.

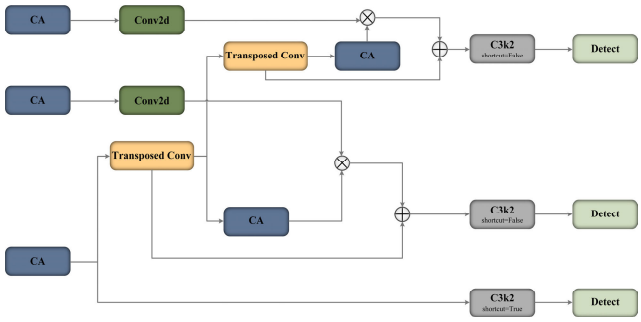


FIGURE 6. Structure of HS-FPN.

D. HS-FPN

In order to improve the recognition precision of traffic targets in the nighttime city with low visibility, this paper adopts HS-FPN [38] to replace the original neck structure. Its network structure is shown in Fig. 6.

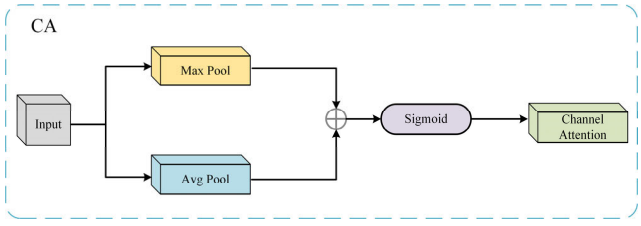


FIGURE 7. The channel attention (CA).

The channel attention (CA) and dimension matching mechanism are used to filter the feature maps of different scales, as shown in Fig. 7. This module effectively extracts the important information in each channel through pooling operation and weight calculation. The function of the feature fusion module is to fuse feature maps of different scales to

enhance the feature representation of the model.

$$MaxPool = Conv2\left(Relu\left(Conv1\left(max_pool(x)\right)\right)\right) \quad (6)$$

$$AvgPool = Conv2\left(Relu\left(Conv1\left(avg_pool(x)\right)\right)\right) \quad (7)$$

$$Conv1 = Conv2d\left(C, \frac{C}{4}\right) \quad (8)$$

$$Conv2 = Conv2d\left(\frac{C}{4}, C\right) \quad (9)$$

$$out = Sigmoid\left(MaxPool + AvgPool\right) \quad (10)$$

The CA module is computed as shown in equation (6) to (10). The input feature map $C\times W\times H$, W where C denotes the number of channels, H denotes the height of the feature map, and W denotes the width of the feature map is first processed. This feature map goes through two pooling layers - global average pooling and global maximum pooling - after which the resulting features are combined together. Subsequently, a Sigmoid activation function is used to determine the weight values for each channel and finally the weights for each channel. Finally, the channel weights are multiplied with the inputs to obtain a feature map with an output of $C\times W\times H$.

By applying HS-FPN as the feature processing structure to fuse C3k2 and other primitive modules belonging to YOLO, the model's ability to face the fusion and extraction of multi-scale features is effectively enhanced, so that the accuracy and robustness of the detection can be improved.

E. FOCALER-GIoU

In order to further improve the detection performance of the model, the Focaler-GIoU [39] loss function is used in this paper. First, the IoU is introduced as a basis and calculated as follows:

$$IoU = \left|\frac{A \cap B}{A \cup B}\right| \quad (11)$$

where A is the prediction frame, B is the real frame, and IoU is the intersection and concurrency ratio between the prediction frame and the real frame.

In the target detection task, IoU can be a good description of how good the target detection model is. The higher the overlap between frame A and frame B, the larger the value of IoU, the better the prediction performance of the model. However, as shown in Fig. 8, when the gradient is zero when there is no intersection between the predicted frame and the real frame, it is impossible to make an accurate judgment on the loss.



FIGURE 8. Non-cross-cutting situations.

The GIoU loss function is presented for optimization in order to address this issue, and it is computed as follows:

$$L_{GIOU} = 1 - IoU + \frac{|C - (A \cup B)|}{|C|} \quad (12)$$

where C is the area of the smallest outer rectangle of the two boxes, as shown in Fig. 9.

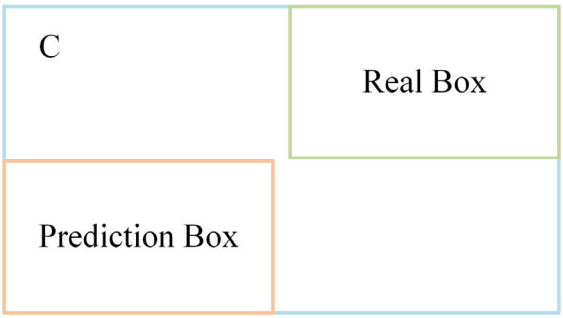


FIGURE 9. Minimum external matrix.

The application of the GIoU loss function characterizes the distance between two boxes by introducing the area of the smallest outer rectangle, which can not only focus on the overlapping region but also focus on other non-overlapping regions and can better respond to the overlap between the two. To continue optimizing edge regression, we propose Focaler-GIoU, a loss function for bounding box regression in target detection. The Focaler-GIoU loss is defined as follows:

$$L_{Focaler-GIoU} = L_{GIoU} + IoU - IoU^{focaler} \quad (13)$$

$$IoU^{focaler} = \begin{cases} 0, & \text{IoU} < d \\ \dfrac{IoU - d}{u - d}, & d \ll IoU \ll u \\ 1, & IoU > u, \end{cases} \quad (14)$$

The problem of sample imbalance is prevalent in target detection tasks. For detection tasks that are dominated by simple samples, the focus should be on simple samples during the bounding box regression process, which helps to improve the detection performance. On the contrary, for the detection that contains a large number of difficult samples, such as small targets, the focus should be on the bounding box regression of the difficult samples, so as to improve the overall precision of target detection. Adjusting the parameters d and u (In this paper d and u are set to 0.05 and 0.95 respectively) in the above equation can enable Focaler-GIoU to focus on different types of regression samples and reconstruct the loss through linear interval mapping, thus realizing the improvement of model detection performance.

In summary, in order to solve the problem that optimization cannot be performed when the gradient disappears due to the value of traditional IOU is 0, GIOU is introduced so that optimization can still be performed when there is no intersection gradient of the two objectives is zero. To further improve the convergence speed of difficult samples, Focaler-GIoU is formed on this basis.

IV. EXPERIMENTAL PROCESS

A. DATASET

In this paper, we use the publicly available dataset named "FLIR" [40]. Because some samples in the FLIR dataset account for too low a proportion and the detection error is large, the original dataset is screened in this paper. The screened dataset includes 10 target types totaling 10,742 images. Among them, the training set, test set, and validation set contain 7519, 1075, and 2148 images, respectively.

B. EXPERIMENTAL DETAILS

All of the experiments in this paper were carried out on a workstation running Ubuntu 22.04. The graphics processor used is a GeForce RTX 3090. And the training and test data are from the public dataset "FLIR.".

To ensure consistency in the training process for both the ablation experiment and comparison tests, we used uniform hyperparameters for training. In particular, the maximum epoch is set to 300, the batch size is 16, and the patience is 100.

C. PERFORMANCE METRICS

In this paper, The performance of the night traffic target detection model was evaluated by using the generalized evaluation criteria. In the evaluation, P indicates the precision of the model. Equations (15) to (16) give the precision formulas for single classification as well as overall, respectively:

$$P_i = \frac{TP}{TP + FP} \tag{15}$$

$$P_{ALL} = \sum_{i=1}^{N} \frac{n_i}{n_{all}} P_i \tag{16}$$

where TP(True Positive) is the true target correctly detected by the model, FP(False Positive) is the target incorrectly reported by the model, N is the total number of categories of the samples10, n_i is the number of samples for each category in the validation set, and n_{all} is the total number of samples in the validation set.

R indicates the percentage of targets that are correctly identified. Equations (17) to (18) give the recall formulas for uniclassification as well as overall, respectively:

$$R_i = \frac{TP}{TP + FN} \tag{17}$$

$$R_{ALL} = \sum_{i=1}^{N} \frac{n_i}{n_{all}} R_i \tag{18}$$

where FN (False Negative) is the real target that the model fails to detect.

Besides, we employ mAP to comprehensively evaluate the precision of target recognition, and use the FSP to measure the inference efficiency of the model. Equations (19) to (21) illustrate how the above indicators are calculated.

$$mAP = \frac{1}{N} \sum_{j=1}^{N} AP_j \tag{19}$$

$$AP_j = \int_0^1 p\,(r)\,dr \tag{20}$$

$$FPS = \frac{1}{t} \tag{21}$$

where AP_j (Average Precision) denotes the model's ability to detect a specific category, $p(r)$ is the Precision-Recall curve generated by the model, and t is the time required to process a single frame.

V. RESULTS

A. COMPARISON WITH THE BASELINE MODEL

In this paper, we suggest a traffic target detection model for nighttime urban environments, YOLOv11-ND, which mainly improves YOLOv11 in the following three aspects to realize the improvement of detection precision and target recall and then improve the safety of nighttime traffic: First, we adopt a more lightweight WTC3k2 module to replace the original C3k2 in the backbone part. Then, the neck part of YOLOv11 is improved using the HS-FPN network structure, which significantly improves the feature extraction capability of the model. Finally, the detection performance of the model is further improved by introducing the Focaler-GIoU loss function. We used the publicly available dataset FLIR for our experimental validation, and the comparison of the improved model YOLOv11-ND with the baseline model YOLOv11 for each classification performance metric is shown in Table 1.

Table 1 shows that YOLOv11-ND increases P, R, mAP50, and mAP50-95 measures by 3.1%, 3.4%, 4%, and 3.6%, respectively, when compared to the YOLOv11 model. However, the FPS decreased by roughly 15 frames per second, while the number of parameters increased somewhat, by about 32w. However, for traffic detection tasks in nighttime environments, a slight decrease in speed is still sufficient to meet the needs, and higher precision is more important.

B. ABLATION EXPERIMENT

In order to verify the effectiveness of the improvement, we used the FLIR dataset for ablation experiments. The following conclusions can be obtained by analyzing the data in Table 2:

The WTC3k2 module performs a wavelet downsampling operation on the input image to obtain a low-frequency map and three high-frequency maps and then improves the corresponding response of the convolutional layer by continuously decomposing the low-frequency maps. Finally, small kernel deep convolution is performed on different frequency maps, and the outputs of different frequencies are merged using IWT. This method effectively increases the sensory field of the input and reduces the number of parameters of the model.

As a result, none of the other metrics of the model were affected, and the number of parameters decreased by about 2%. This demonstrates that the WTC3k2 module achieves a lightweight model while maintaining precision.

The main function of the HS-FPN module is to effectively combine high-level feature semantic information and

TABLE 1. Units for magnetic properties.

Categorization	Modle	P	R	mAP50	mAP50-90	parameters	FPS
All	YOLOv11	0.793	0.602	0.691	0.423	2585272	83
	YOLOv11-ND	0.824	0.636	0.731	0.459	2904080	68
Person	YOLOv11	0.813	0.702	0.792	0.464		
	YOLOv11-ND	0.847	0.725	0.819	0.500		
Bike	YOLOv11	0.748	0.589	0.652	0.346		
	YOLOv11-ND	0.784	0.623	0.707	0.399		
Car	YOLOv11	0.803	0.735	0.810	0.558		
	YOLOv11-ND	0.836	0.762	0.837	0.586		
Motor	YOLOv11	0.792	0.678	0.737	0.424		
	YOLOv11-ND	0.871	0.702	0.792	0.454		
Bus	YOLOv11	0.778	0.742	0.791	0.557		
	YOLOv11-ND	0.741	0.710	0.796	0.581		
Trunk	YOLOv11	0.588	0.578	0.571	0.398		
	YOLOv11-ND	0.640	0.614	0.612	0.402		
Light	YOLOv11	0.856	0.476	0.610	0.308		
	YOLOv11-ND	0.865	0.519	0.653	0.339		
Hydrant	YOLOv11	0.755	0.275	0.439	0.213		
	YOLOv11-ND	0.739	0.303	0.469	0.236		
Sign	YOLOv11	0.788	0.406	0.517	0.309		
	YOLOv11-ND	0.838	0.473	0.584	0.360		
Other vehicle	YOLOv11	0.678	0.508	0.536	0.298		
	YOLOv11-ND	0.671	0.539	0.589	0.330		

TABLE 2. Results of ablation experiment.

Model			P	R	mAP50	mAP50-90	parameters	FPS
WTC3k2	HS-FPN	Focaler-GIoU						
×	×	×	0.793	0.602	0.691	0.423	2585272	83
√	√	×	0.826	0.626	0.725	0.454	2904080	68
√	×	√	0.787	0.616	0.691	0.421	2525920	80
×	√	√	0.822	0.623	0.729	0.458	2963432	66
√	√	√	0.824	0.636	0.731	0.459	2904080	68

TABLE 3. Results of comparison with other models.

Model	P	R	mAP50	mAP50-95	Parameters	FPS
YOLOv3	0.816	0.593	0.689	0.424	4056920	67
YOLOv5	0.777	0.607	0.690	0.420	2506064	85
YOLOv6	0.782	0.571	0.654	0.396	4233843	86
YOLOv8	0.799	0.607	0.699	0.425	3008768	79
YOLOv9	0.800	0.590	0.679	0.418	1732944	70
YOLOv10	0.754	0.623	0.697	0.426	2700656	72
YOLOv11	0.793	0.602	0.691	0.423	2585272	83
RT-DETR	0.624	0.503	0.506	0.372	16768924	82
RT-DETR-ResNet50	0.658	0.544	0.591	0.322	31491020	61
Ours	0.824	0.636	0.731	0.459	2904080	68

low-scale feature detail information through the application of the feature selection module and feature fusion module, so as to improve the ability of the model to deal with the fusion and extraction of multi-scale features. As can be seen from the experimental data, except for the slight rise in the number of model parameters, all other indexes have been significantly improved.

The application of the Focaler-GIoU loss function allows the model to focus on different types of regression samples and reconstruct the loss through linear interval mapping, thus realizing the improvement of detection performance. The experimental data show that the Focaler-GIoU loss function improves the recall of the model by about 1.6%, enhances the target detection ability under low illumination conditions,

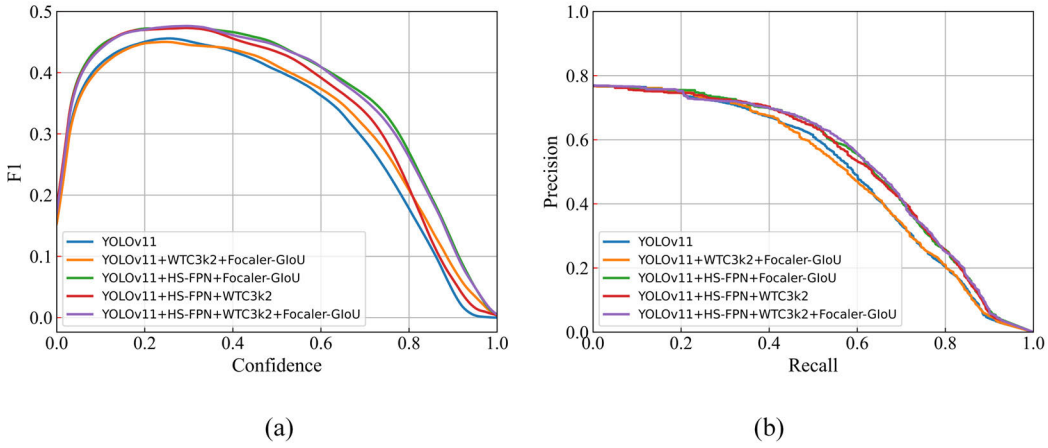


FIGURE 10. Results of Ablation experiment. (a) F1 curve; (b) PR curve.

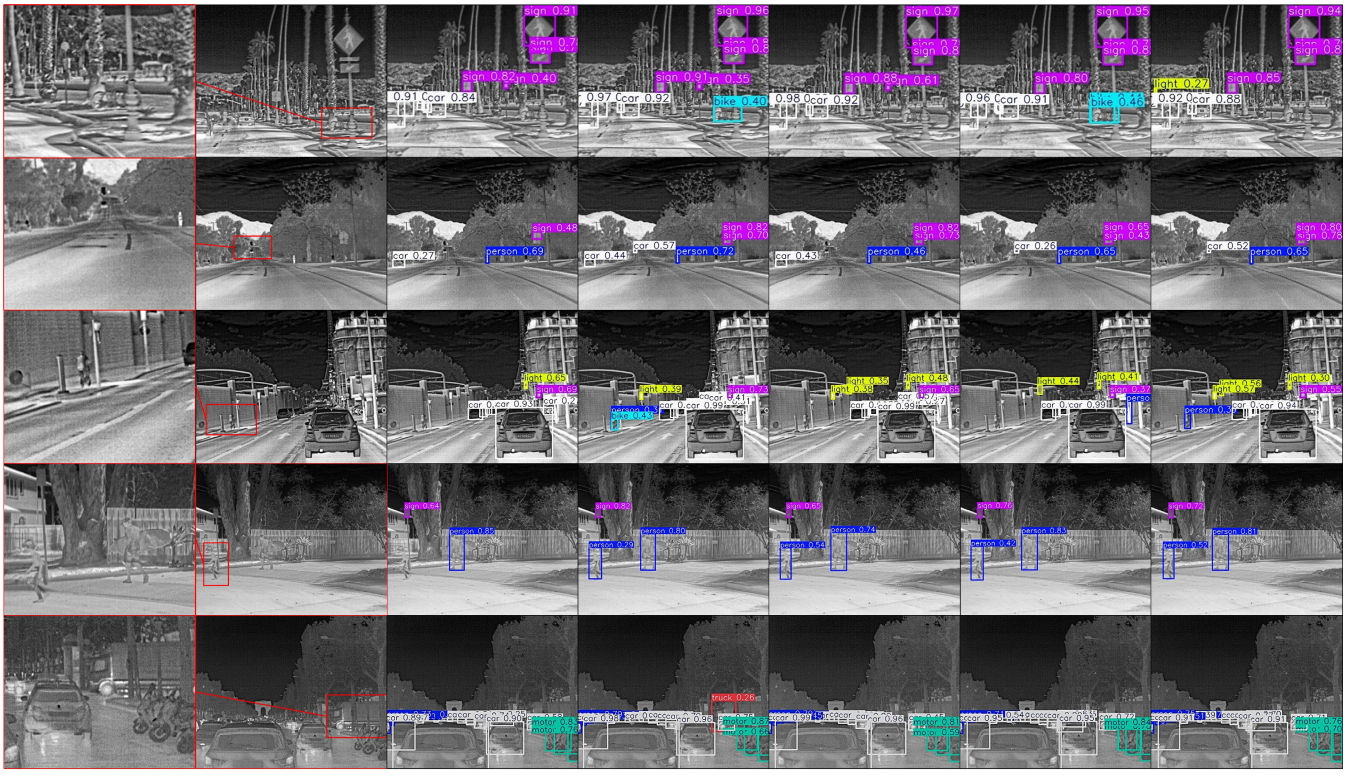


FIGURE 11. Examples of ablation test comparisons. (a) The original image; (b) YOLOv11; (c) YOLOv11+WTC3k2+HS-FPN+Focaler-GIoU; (d) YOLOv11+HS-FPN+Focaler-GIoU; (e) YOLOv11+WTC3k2+Focaler-GIoU; (f) YOLOv11+ WTC3k2+HS-FPN.

and the mAP50 and mAP50-90 are also slightly improved. Although the detection precision P decreased, it was only 0.2%.

Fig. 10 shows the comparison of the results of the F1 curve and PR curve in the ablation experiment, which shows that the improved algorithm curve completely surrounds the model curve of YOLOv11. It can be concluded that the improved algorithm YOLOv11-ND has better detection performance in nighttime urban environments.

Fig. 11 exhibits the comparative results of the ablation experiments in different scenarios, and the improvement effect of the nighttime detection model YOLOv11-ND compared to the baseline model YOLOv11 can be clearly seen. In the complex nighttime urban environment, the improved

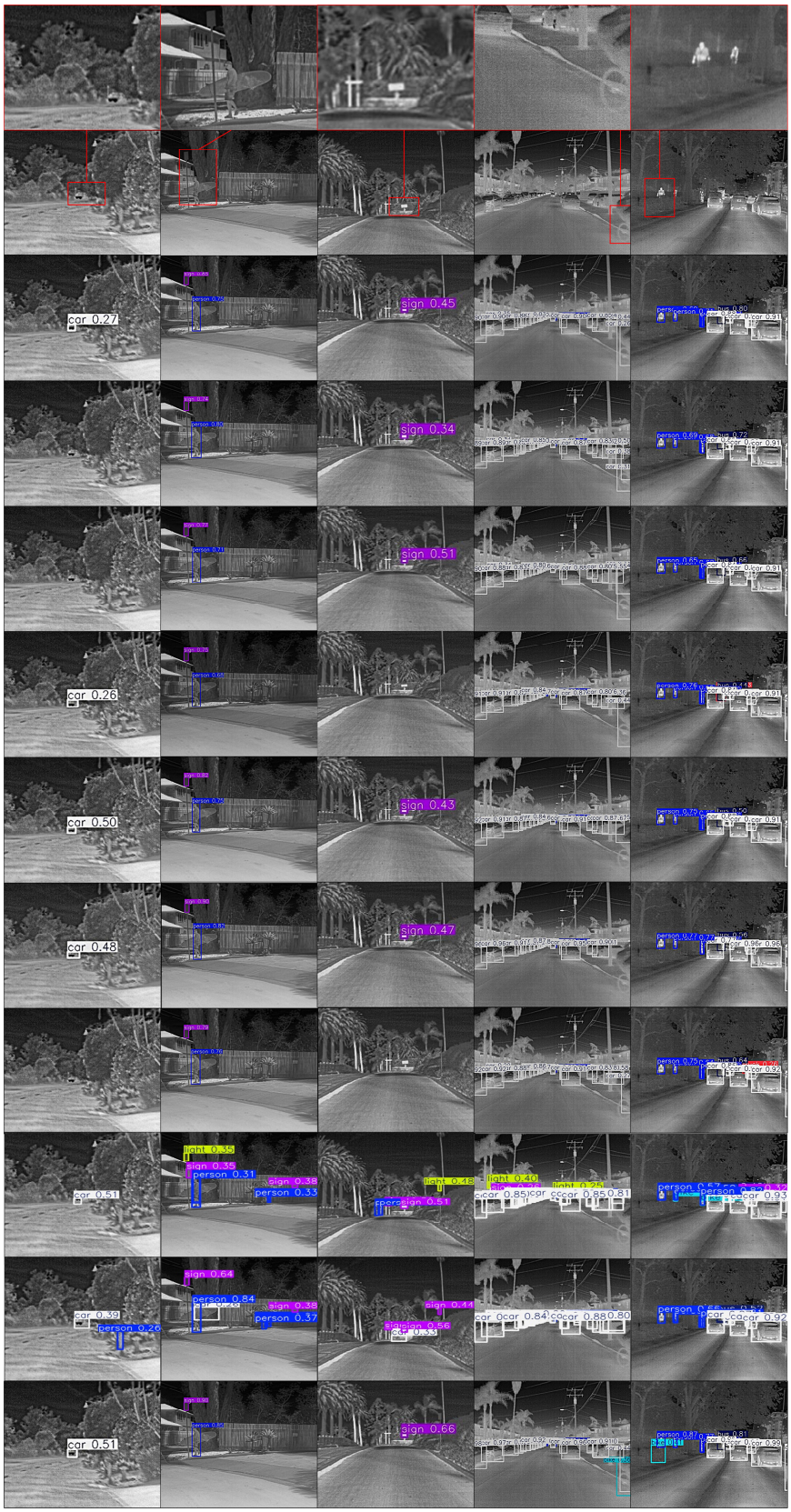


FIGURE 12. Comparison results with other models. (a) The original image; (b) YOLOv3; (c) YOLOv5; (d) YOLOv6; (e) YOLOv8; (f) YOLOv9; (g) YOLOv10; (h) YOLOv11; (i) RT-DETR;(j) RT-DETR-ResNet50;(k) Ours.

model has fewer missed and false detections compared to the baseline model, YOLOv11. And the confidence level of YOLOv11-ND detection is higher when detecting the same target.

C. COMPARISON WITH OTHER MODELS

To further verify the effectiveness of the improvement, we compare the results of the improved algorithm with the following detection algorithms: (a) YOLOv3; (b) YOLOv5; (c) YOLOv6; (d) YOLOv8; (e) YOLOv9; (f) YOLOv10; (g) RT-DETR; (h) RT-DETR-ResNet50.The experimental results are shown in Table 3, and the precision-related performance metrics of YOLOv11-ND are also ahead of other models. performance indexes are also ahead of other models.

Fig. 12 shows the comparison results with other models in different scenes. It can be seen that our improved model, YOLOv11-ND, has a lower leakage rate and a higher confidence level in a low-illumination night environment. And in the complex environment with more occlusion, our model has a lower false detection rate. It can be seen that the improved model has better performance for the detection of traffic targets in urban environments at night.

VI. CONCLUSION

The paper suggests an enhanced nighttime detection model based on YOLOv11—YOLOv11-ND to address the issues of challenging traffic target detection and low recognition precision in the evening urban environment. YOLOv11-ND adopts the WTC3k2 module to replace the original C3k2 module in the backbone part, which reduces the number of parameters of the model while ensuring precision. In order to increase the model's ability to handle the fusion and extraction of multi-scale features, we then use the HS-FPN network structure, which has a greater feature extraction ability in the neck part. This results in a notable improvement in the model performance index. Lastly, to further improve the model's detection performance, the Focaler-GIoU loss function is shown.

In comparison to YOLOv11, YOLOv11-ND improves the P, R, mAP50, and mAP50-95 measures by 3.1%, 3.4%, 4%, and 3.6%, respectively, according to our experimental results on the FLIR dataset. However, the FPS decreases by roughly 15 frames per second, and the number of parameters increases a little, by about 32w. Regarding precision performance, the model demonstrates superior detection capabilities for larger, well-defined targets such as person and bikes. Conversely, its performance degrades for smaller targets like hydrants, and for categories with insufficient sample representation in the dataset, such as buses and other vehicles. In summary, the model exhibits certain limitations in detecting small targets and categories with limited samples, coupled with a slight reduction in inference speed. Nevertheless, for nighttime traffic detection tasks, the marginally decreased speed remains acceptable, while the higher overall detection accuracy is of greater significance. This indicates that YOLOv11-TD is better suited for nighttime traffic object detection in urban environments.

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